

SIT283 Assignment 1: Conveyor Belt

Goal:

“Create a virtual environment with a conveyor belt of different objects that have to be sorted by hand into different containers.

Focus on adding objects to the belt, moving them along the belt, and allowing them to be manipulated by the controller.”

Github: <https://github.com/RyanSpeirs/SIT283-Week-1>

Method:

The simplest part of this was what I tackled first, the conveyor belt. The “Trampoline” demonstrated in the workshop was a suitable starting point, the code isn’t elegant but I just removed the randomisation, set the z axis to have the most minimal velocity, and changed the condition from `OnCollisionEnter`, to `OnCollisionStay` as that means constant application of force. This does have an issue that objects would accelerate until they leave the conveyor, hence why the values were dialled down.

Setting up the items to be interactable was probably the easiest part, just apply the same stuff as the sample interactable. The real challenge was configuring the XR elements, as Unity 6’s default setting is to bulk load a lot of the stuff built-in with XR plugins and this caused crashes and a lot of junk data to accrue. I made the items and the bins myself in blender as quirky unique objects that stand out by silhouette and colour.

The Bins and Spawner were where I ran into some trouble. The physics aspect was fine, but the problem stemmed from modelling a system that can identify when the wrong item is in the wrong bin. The spawner is also meant to stop spawning an item type once it’s bin is full. I tried using Unity Events as I like the idea of wiring in the inspector instead of hard-coding references, but of the 3 events, it was the less important UI events that worked fine, while the Spawner was less successful. I had an initial configuration where the spawner would track item counts but after running into issues where it would refuse to spawn items, even after consulting Claude AI, I pivoted to a model where the bins would remember count. That is why there is notably a large amount of debug logging in these scripts.

Testing this app was probably the most stressful. Due to constraints on my circumstances, I’m limited to a laptop with the only variant of a GTX GPU that isn’t supported by Air Link, so I’ve been testing everything that isn’t entirely XR in the editor, and only pushing builds to my headset for checking interaction and world display.

AI Usage:

Claude AI - Sonnet 5; Medium

“Lets work with a script I have called ItemBin; its going onto 3 prefabs, it needs to allow me to pick an item in the inspector and it counts that item upto 10, and at ten notifies the spawner to stop with that item, it also needs to update a UI script I don't have yet “

“the spawner exists already “

“the spawner just has a limit value that applies to each type. “

```
public class ItemSpawner : MonoBehaviour
{
    [System.Serializable]
    public class ItemTypeConfig
    {
        public string typeName = "Item";
        public GameObject prefab;
        public int targetCount = 10;

        [HideInInspector] public int spawnedCount = 0;
    }

    [Header("Item Types")]
    public ItemTypeConfig[] itemTypes = new ItemTypeConfig[3];

    [Header("Spawn Settings")]
    public Transform spawnPoint;
    public float spawnInterval = 2f;
    public bool autoStart = true;

    private bool isSpawning = false;

    private void Start()
    {
        if (spawnPoint == null) spawnPoint = transform;

        if (autoStart) StartSpawning();
    }

    public void StartSpawning()
    {
        if (isSpawning) return;
        isSpawning = true;
        StartCoroutine(SpawnRoutine());
    }

    public void StopSpawning()
```

```

{
    isSpawning = false;
    StopAllCoroutines();
}

private IEnumerator SpawnRoutine()
{
    while (isSpawning)
    {
        List<ItemTypeConfig> available = GetAvailableTypes();

        if (available.Count == 0)
        {
            Debug.Log("[ItemSpawner] All item type targets reached. Stopping spawner.");
            isSpawning = false;
            yield break;
        }

        ItemTypeConfig chosen = available[Random.Range(0, available.Count)];
        SpawnItem(chosen);

        yield return new WaitForSeconds(spawnInterval);
    }
}

private List<ItemTypeConfig> GetAvailableTypes()
{
    List<ItemTypeConfig> result = new List<ItemTypeConfig>();
    foreach (var type in itemTypes)
    {
        if (type.prefab != null && type.spawnedCount < type.targetCount)
            result.Add(type);
    }
    return result;
}

private void SpawnItem(ItemTypeConfig type)
{
    Instantiate(type.prefab, spawnPoint.position, spawnPoint.rotation);
    type.spawnedCount++;
    Debug.Log($"[ItemSpawner] Spawned {type.typeName}
({type.spawnedCount}/{type.targetCount})");
}
}

```

“So I have 3 item prefabs with the item script on them, and I just edit the string value to match the item name, then alter bin. but spawner looks for them and knows them differently. I want to standardise how every thing "knows" what item is what. “

“So my 3 items are a cyan cube with flattened corners, a pair of magenta diamonds that are joined at one tip, with the outer tips flattened and the middle flattened, and a yellow shape that is two cones joined at the tips. Magenta and yellow tend to roll a lot. “

“I have 3 TMP text objects: YellowCounter MagentaCounter and CyanCounter for the items and a 4th JobDoneText for my UI, can you check this

```
using TMPro;
using UnityEngine;
```

```
public class UIManager : MonoBehaviour
{
    public TMP_Text cyanCounter;
    public TMP_Text magentaCounter;
    public TMP_Text yellowCounter;
    public TMP_Text jobDoneText;

    private bool cyanFull = false;
    private bool magentaFull = false;
    private bool yellowFull = false;

    private void Start()
    {
        if (jobDoneText != null) jobDoneText.gameObject.SetActive(false);
    }

    // This updates the UI
    public void UpdateCount(ItemType type, int count)
    {
        switch (type)
        {
            case ItemType.CyanCube:
                if (cyanCounter != null) cyanCounter.text = $"Cyan: {count}/5";
                break;
            case ItemType.Magenta:
                if (magentaCounter != null) magentaCounter.text = $"Magenta: {count}/5";
                break;
            case ItemType.Yellow:
                if (yellowCounter != null) yellowCounter.text = $"Yellow: {count}/5";
                break;
        }
    }

    // When all 3 bins are full, displays Job Done message
```

```
public void NotifyBinFull(ItemType type)
{
    switch (type)
    {
        case ItemType.CyanCube: cyanFull = true; break;
        case ItemType.Magenta: magentaFull = true; break;
        case ItemType.Yellow: yellowFull = true; break;
    }

    if (cyanFull && magentaFull && yellowFull)
    {
        if (jobDoneText != null) jobDoneText.gameObject.SetActive(true);
    }
}
}
```

“yes this is on canvas and all text objects are a child of the object hosting the UI script”

“nothing is spawning now: all 4 spawners

[ItemSpawner] All item type targets reached. Stopping spawner. UnityEngine.Debug:Log (object) ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:65) UnityEngine.MonoBehaviour:StartCoroutine (System.Collections.IEnumerator) ItemSpawner:StartSpawning () (at Assets/Scripts/ItemSpawner.cs:38) ItemSpawner:Start () (at Assets/Scripts/ItemSpawner.cs:31) [ItemSpawner] All item type targets reached. Stopping spawner. UnityEngine.Debug:Log (object) ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:65) UnityEngine.MonoBehaviour:StartCoroutine (System.Collections.IEnumerator) ItemSpawner:StartSpawning () (at Assets/Scripts/ItemSpawner.cs:38) ItemSpawner:Start () (at Assets/Scripts/ItemSpawner.cs:31) [ItemSpawner] All item type targets reached. Stopping spawner. UnityEngine.Debug:Log (object) ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:65) UnityEngine.MonoBehaviour:StartCoroutine (System.Collections.IEnumerator) ItemSpawner:StartSpawning () (at Assets/Scripts/ItemSpawner.cs:38) ItemSpawner:Start () (at Assets/Scripts/ItemSpawner.cs:31) [ItemSpawner] All item type targets reached. Stopping spawner. UnityEngine.Debug:Log (object) ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:65) UnityEngine.MonoBehaviour:StartCoroutine (System.Collections.IEnumerator) ItemSpawner:StartSpawning () (at Assets/Scripts/ItemSpawner.cs:38) ItemSpawner:Start () (at Assets/Scripts/ItemSpawner.cs:31)

“I can't move in my VR environment, how do I configure controls to teleport and grab things?”

"I have pushed a build to the headset but it displays as a flat panel, I can't move the camera, or interact with things"

"Could not allocate memory: System out of memory! Trying to allocate: 24390242B with 16 alignment. MemoryLabel: DynamicArray Allocation happened at: Line:163 in C:\build\output\unity\unity\Modules\NativeKernel\Include\NativeKernel\Core\Containers\Allocators.h Memory overview [ALLOC_MEMORYPROFILER] used: 1699864B &s| reserved: 477912522928B [ALLOC_TEMP_TLS] used: 8B &s| reserved: 477912522928B [ALLOC_DEFAULT] used: 4088856684B &s| reserved: 477912522928B [ALLOC_TEMP_JOB_1_FRAME] used: 0B &s| reser... (For the full message, see the editor log file) "

"Can you tell me what is causing this?"

"Could not allocate memory: System out of memory! Trying to allocate: 24390242B with 16 alignment. MemoryLabel: DynamicArray Allocation happened at: Line:163 in C:\build\output\unity\unity\Modules\NativeKernel\Include\NativeKernel\Core\Containers\Allocators.h Memory overview [ALLOC_MEMORYPROFILER] used: 1426832B &s| reserved: 376980821760B [ALLOC_TEMP_TLS] used: 0B &s| reserved: 376980821760B [ALLOC_DEFAULT] used: 5341985284B &s| reserved: 376980821760B [ALLOC_TEMP_JOB_1_FRAME] used: 0B &s| reser... (For the full message, see the editor log file)"

"I haven't done any tests though, I just re-opened the editor and went into XR Plug-in management to fix some yellow warnings "

"Could not allocate memory: System out of memory! Trying to allocate: 24390242B with 16 alignment. MemoryLabel: DynamicArray Allocation happened at: Line:163 in C:\build\output\unity\unity\Modules\NativeKernel\Include\NativeKernel\Core\Containers\Allocators.h Memory overview [ALLOC_MEMORYPROFILER] used: 1426832B &s| reserved: 376980821760B [ALLOC_TEMP_TLS] used: 0B &s| reserved: 376980821760B [ALLOC_DEFAULT] used: 5341985284B &s| reserved: 376980821760B [ALLOC_TEMP_JOB_1_FRAME] used: 0B &s| reser... (For the full message, see the editor log file)"

"water fox has 700MB with only 3 tabs open, this and two unity documentation pages "

"Do I just restart the computer to clean-up any thing hanging then? I have 800 MB in water fox, antimalware is 180 MB, hub is 95 and task manager is 100 steady, there is also mixed REality Link but thats 17 MB "

"Could not allocate memory: System out of memory! Trying to allocate: 24390242B with 16 alignment. MemoryLabel: DynamicArray Allocation happened at: Line:163 in C:\build\output\unity\unity\Modules\NativeKernel\Include\NativeKernel\Core\Containers\Allocators.h Memory overview [ALLOC_MEMORYPROFILER] used: 1426832B &s| reserved: 376980821760B [ALLOC_TEMP_TLS] used: 0B &s| reserved: 376980821760B [ALLOC_DEFAULT] used: 5341985284B &s| reserved: 376980821760B [ALLOC_TEMP_JOB_1_FRAME] used: 0B &s| reser... (For the full message, see the editor

log file)”

“I can move my position, and its still loading as a 2D screen. “

“Alright, so things are some what there. interaction works in the build but its limited, the UI doesn't display. I've been putting off an issue where the spawner just spits out roughly 10 objects and then stops “

“no I have a big red overlay effect to advise of play mode editing. the session started and all 4 reported they'd hit the limit.”

“I had a better idea I've shifted the counting onto the bins, can you check spawner first

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
```

```
public class ItemSpawner : MonoBehaviour
{
    [System.Serializable]
    public class ItemTypeConfig
    {
        public ItemType type;
        public GameObject prefab;

        public bool stopped = false;
    }

    [Header("Item Types")]
    public ItemTypeConfig[] itemTypes = new ItemTypeConfig[3];

    [Header("Spawn Settings")]
    public Transform spawnPoint;
    public float spawnInterval = 2f;
    public bool autoStart = true;

    private bool isSpawning = false;

    private void Start()
    {
        if (spawnPoint == null) spawnPoint = transform;
        if (autoStart) StartSpawning();
    }

    public void StartSpawning()
    {

```

```

        if (isSpawning) return;
        isSpawning = true;
        StartCoroutine(SpawnRoutine());
    }

    public void StopSpawning()
    {
        isSpawning = false;
        StopAllCoroutines();
    }

    // remember to wire bins to this
    public void ForceStopType(ItemType type)
    {
        foreach (var t in itemTypes)
        {
            if (t.type == type)
                t.stopped = true;
        }
    }

    private IEnumerator SpawnRoutine()
    {
        while (isSpawning)
        {
            List<ItemTypeConfig> available = GetAvailableTypes();

            if (available.Count == 0)
            {
                Debug.Log("[ItemSpawner] All item types stopped. Stopping spawner.");
                isSpawning = false;
                yield break;
            }

            ItemTypeConfig chosen = available[Random.Range(0, available.Count)];
            SpawnItem(chosen);

            yield return new WaitForSeconds(spawnInterval);
        }
    }

    private List<ItemTypeConfig> GetAvailableTypes()
    {
        List<ItemTypeConfig> result = new List<ItemTypeConfig>();
        foreach (var type in itemTypes)
        {
            if (type.prefab != null && !type.stopped)
                result.Add(type);
        }
    }

```



```
    }  
    return result;  
}  
  
private void SpawnItem(ItemTypeConfig type)  
{  
    GameObject obj = Instantiate(type.prefab, spawnPoint.position, spawnPoint.rotation);  
  
    Item item = obj.GetComponent<Item>();  
    if (item != null) item.type = type.type;  
  
    Debug.Log($"[ItemSpawner] Spawned {type.type}");  
}  
}
```

“lets just tune bin now to make sure it works

```
public class ItemBin : MonoBehaviour  
{  
    [SerializeField] private ItemType acceptedItem;  
    [SerializeField] private int capacity = 10;  
  
    public ItemTypeIntEvent OnCountChanged;  
    public ItemTypeEvent OnBinFull;  
  
    private int currentCount = 0;  
  
    private void OnTriggerEnter(Collider other)  
    {  
  
        Item item = other.GetComponent<Item>();  
        if (item == null) return; // we only care about Items  
  
        if (item.type != acceptedItem) // wrong items get destroyed  
        {  
            Destroy(other.gameObject);  
            return;  
        }  
  
        if (currentCount >= capacity) return; // If bin is full, stop accepting items  
  
        currentCount++;  
        OnCountChanged?.Invoke(acceptedItem, currentCount);  
        Destroy(other.gameObject);  
  
        if (currentCount >= capacity) // tells the spawner and UI the bin is full  
            OnBinFull?.Invoke(acceptedItem);  
    }  
}
```

}”

“UI still isn’t showing”

“UI is set to world space but not showing “

“I tried your suggestions, its far off in the distance and bigger than god “

“Its not sitting as a UI on camera so I have it as a billboard like display. Now I got to 10 cyan cubes and it continued spawning them “

“Ok so I set the count to 5 and it still produced 3 cyan cubes after hitting it, same with the other items, and no Job Done display”

“how do I check logs from the headset?”

“I am sorting the items into the bins and there was quite a gap before it produced the excess cubes “

“no they were spawned 8, 10 and 14 seconds after the bin was full “

“I can’t see the hierarchy as I can only test via builds in headset, so no logs either “

“there is only one spawner. The issue is that either the bin full message isn't sending, or the stop type method doesn't register “

“adb is set up as I use Mixed Reality Link and that requires ADB drivers “

“by positioning the spawner over each bin, I got them all to 5, which should have ended spawning, but it kept spawning, also the Job done! text did not spawn “

“no log entries for ForceStopType, so its not being called ever”

“Loading OpenXR loader library at path: C:\Users\Ryan\Documents\SIT\SIT283\SIT283 Week 1\Library\PackageCache\com.unity.xr.openxr@ef0033a586bf\RuntimeLoaders\windows\x64\openxr_loader.dll
UnityEngine.XR.Management.XRGeneralSettings:AttemptInitializeXRSDKOnLoad () (at ./Library/PackageCache/com.unity.xr.management@e3a3882b360a/Runtime/XRGeneralSettings.cs:145) XR: Error setting active audio output driver. Falling back to default.
UnityEngine.XR.Management.XRGeneralSettings:AttemptInitializeXRSDKOnLoad () XR: Error setting active audio output driver. Falling back to default.
UnityEngine.XR.Management.XRGeneralSettings:AttemptInitializeXRSDKOnLoad () No ray and select interactor found!
UnityEngine.XR.Interaction.Toolkit.Gaze.XRGazeAssistance:Start () (at ./Library/PackageCache/com.unity.xr.interaction.toolkit@8a8cfde33e41/Runtime/Interaction/Gaze/XRGazeAssistance.cs:413) No ray and select interactor found!
UnityEngine.XR.Interaction.Toolkit.Gaze.XRGazeAssistance:Start () (at

```
./Library/PackageCache/com.unity.xr.interaction.toolkit@8a8cfde33e41/Runtime/Interaction/
Gaze/XRGazeAssistance.cs:413) [ItemSpawner] All item types stopped. Stopping spawner.
UnityEngine.Debug:Log (object) ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at
Assets/Scripts/ItemSpawner.cs:65) UnityEngine.MonoBehaviour:StartCoroutine
(System.Collections.IEnumerator) ItemSpawner:StartSpawning () (at
Assets/Scripts/ItemSpawner.cs:37) ItemSpawner:Start () (at
Assets/Scripts/ItemSpawner.cs:30) Both Ray Interactor and Near-Far Interactor are
assigned. Only one should be assigned, not both. Clearing Ray Interactor.
UnityEngine.InputSystem.LowLevel.NativeInputRuntime/<>c__DisplayClass7_0:<set_onUpd
ate>b__0
(UnityEngineInternal.Input.NativeInputUpdateType,UnityEngineInternal.Input.NativeInputEve
ntBuffer*) UnityEngineInternal.Input.NativeInputSystem:NotifyUpdate
(UnityEngineInternal.Input.NativeInputUpdateType,intptr)
UnityEngine.GUIUtility:ProcessEvent (int,intptr,bool&) Both Ray Interactor and Near-Far
Interactor are assigned. Only one should be assigned, not both. Clearing Ray Interactor.
UnityEngine.InputSystem.LowLevel.NativeInputRuntime/<>c__DisplayClass7_0:<set_onUpd
ate>b__0
(UnityEngineInternal.Input.NativeInputUpdateType,UnityEngineInternal.Input.NativeInputEve
ntBuffer*) UnityEngineInternal.Input.NativeInputSystem:NotifyUpdate
(UnityEngineInternal.Input.NativeInputUpdateType,intptr)
UnityEngine.GUIUtility:ProcessEvent (int,intptr,bool&) “
```

```
“ItemSpawner:SpawnItem (ItemSpawner/ItemTypeConfig) (at
Assets/Scripts/ItemSpawner.cs:95)
ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:71)
UnityEngine.SetupCoroutine:InvokeMoveNext (System.Collections.IEnumerator,intptr)
[ItemBin] Wrong type (CyanCube, expected Magenta) — destroying
UnityEngine.Debug:Log (object)
ItemBin:OnTriggerEnter (UnityEngine.Collider) (at Assets/Scripts/ItemBin.cs:25)
[ItemSpawner] Spawned Magenta
UnityEngine.Debug:Log (object)
ItemSpawner:SpawnItem (ItemSpawner/ItemTypeConfig) (at
Assets/Scripts/ItemSpawner.cs:95)
ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:71)
UnityEngine.SetupCoroutine:InvokeMoveNext (System.Collections.IEnumerator,intptr)
[ItemBin] Counted! Magenta now at 3/5
UnityEngine.Debug:Log (object)
ItemBin:OnTriggerEnter (UnityEngine.Collider) (at Assets/Scripts/ItemBin.cs:37)
[ItemSpawner] Spawned Magenta
UnityEngine.Debug:Log (object)
ItemSpawner:SpawnItem (ItemSpawner/ItemTypeConfig) (at
Assets/Scripts/ItemSpawner.cs:95)
ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:71)
UnityEngine.SetupCoroutine:InvokeMoveNext (System.Collections.IEnumerator,intptr)
[ItemBin] Counted! Magenta now at 4/5
UnityEngine.Debug:Log (object)
ItemBin:OnTriggerEnter (UnityEngine.Collider) (at Assets/Scripts/ItemBin.cs:37)
[ItemSpawner] Spawned Magenta
```

```
UnityEngine.Debug:Log (object)
ItemSpawner:SpawnItem (ItemSpawner/ItemTypeConfig) (at
Assets/Scripts/ItemSpawner.cs:95)
ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:71)
UnityEngine.SetupCoroutine:InvokeMoveNext (System.Collections.IEnumerator,intptr)
[ItemBin] Counted! Magenta now at 5/5
UnityEngine.Debug:Log (object)
ItemBin:OnTriggerEnter (UnityEngine.Collider) (at Assets/Scripts/ItemBin.cs:37)
[ItemBin] Magenta bin full — invoking OnBinFull
UnityEngine.Debug:Log (object)
ItemBin:OnTriggerEnter (UnityEngine.Collider) (at Assets/Scripts/ItemBin.cs:43)
ForceStopType called for Magenta
UnityEngine.Debug:Log (object)
ItemSpawner:ForceStopType (ItemType) (at Assets/Scripts/ItemSpawner.cs:49)
UnityEngine.Events.UnityEvent`1<ItemType>:Invoke (ItemType)
ItemBin:OnTriggerEnter (UnityEngine.Collider) (at Assets/Scripts/ItemBin.cs:44)
[ItemSpawner] Spawned CyanCube
UnityEngine.Debug:Log (object)
ItemSpawner:SpawnItem (ItemSpawner/ItemTypeConfig) (at
Assets/Scripts/ItemSpawner.cs:95)
ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:71)
UnityEngine.SetupCoroutine:InvokeMoveNext (System.Collections.IEnumerator,intptr)
[ItemBin] Wrong type (CyanCube, expected Magenta) — destroying
UnityEngine.Debug:Log (object)
ItemBin:OnTriggerEnter (UnityEngine.Collider) (at Assets/Scripts/ItemBin.cs:25)
[ItemSpawner] Spawned Yellow
UnityEngine.Debug:Log (object)
ItemSpawner:SpawnItem (ItemSpawner/ItemTypeConfig) (at
Assets/Scripts/ItemSpawner.cs:95)
ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:71)
UnityEngine.SetupCoroutine:InvokeMoveNext (System.Collections.IEnumerator,intptr)
[ItemBin] Wrong type (Yellow, expected Magenta) — destroying
UnityEngine.Debug:Log (object)
ItemBin:OnTriggerEnter (UnityEngine.Collider) (at Assets/Scripts/ItemBin.cs:25)
[ItemSpawner] Spawned CyanCube
UnityEngine.Debug:Log (object)
ItemSpawner:SpawnItem (ItemSpawner/ItemTypeConfig) (at
Assets/Scripts/ItemSpawner.cs:95)
ItemSpawner/<SpawnRoutine>d__10:MoveNext () (at Assets/Scripts/ItemSpawner.cs:71)
UnityEngine.SetupCoroutine:InvokeMoveNext (System.Collections.IEnumerator,intptr)
[ItemBin] Wrong type (CyanCube, expected Magenta) — destroying
UnityEngine.Debug:Log (object)
ItemBin:OnTriggerEnter (UnityEngine.Collider) (at Assets/Scripts/ItemBin.cs:25)
“
```

“Bins weren't linked. Spawner still an issue”

“It doesn't matter, the goal is movement of objects and their manipulation by the conveyor

and the player, the bins just add context, I think I've spent too much time polishing the silverware on this one"